

Professional Summary

Full-stack Software Engineer specializing in designing and delivering AI-powered, cloud-native systems for enterprise-scale workloads. Expertise in building distributed microservices with **Java/Spring Boot and .NET (C#)** and developing intuitive front-end solutions using **React, TypeScript, and Power Apps Portals**. Strong focus on **AI/ML-driven automation, MLOps, and real-time inference pipelines** using TensorFlow, Azure ML, and LangChain-based RAG systems.

Recent experience includes designing intelligent enterprise applications integrating **.NET APIs, Dynamics 365 Customer Service, Power Apps Portals, and Microsoft Agent SDK**, automating workflows and enhancing operational efficiency in enterprise environments.

Proven success in:

- Modernizing monolithic systems into scalable event-driven architectures using Kafka and Redis.
- Deploying high-availability and secure applications across AWS, Azure, and GCP.
- Integrating ML models into production using TensorFlow Serving, SageMaker, and FastAPI.
- Driving DevOps automation (Docker, Kubernetes, GitHub Actions, Jenkins) for rapid delivery.

Passionate about building inclusive, data-driven technology that enhances learning, accessibility, and decision-making across domains such as education, analytics, and intelligent enterprise platforms.

Technical Skills

Programming Languages: Java (17+), TypeScript, JavaScript (ES6+), Python, SQL, Bash, **C# (.NET Core, ASP.NET)**.

Backend Frameworks & Libraries: Spring Boot, Spring Cloud, Hibernate, Node.js, Express.js, GraphQL (Netflix DGS), gRPC, JUnit, Mockito, REST Assured, OpenAPI/Swagger, **Microsoft Agent SDK**.

Frontend Technologies: React.js, Angular, HTML5, CSS3, TypeScript, JSON, Responsive Web Design, Accessibility (WCAG 2.1), Cross-Browser Optimization, Server-Side Rendering (SSR), SEO-Driven Architecture, **UI Apps Portals**.

Architecture & Microservices: Distributed Microservices with Spring Boot, API Gateway, Containerized Deployments, Event-Driven Systems (Kafka, AWS Lambda).

APIs & Integrations: RESTful API Development, GraphQL Schema Composition, OAuth 2.0 & JWT Security, Secure API Design, Payment Gateway Integrations, Third-Party Service Orchestration, **Dynamics 365 API Integrations**.

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis; proficient in data modeling, indexing, and query optimization.

AI & Intelligent Systems: TensorFlow Serving, FastAPI-based Model Inference, LangChain Pipelines, NLP API Integrations, Real-Time Prediction Services with gRPC and Python Backends.

Cloud & Infrastructure: AWS (EKS, EC2, S3, RDS, Lambda, CloudFront, IAM), Azure (App Service, **Data Factory, Logic Apps, ML, DevOps**), GCP (Cloud Run, Vertex AI); strong in multi-cloud deployment, scaling, and monitoring.

DevOps & Automation: Jenkins, GitHub Actions, AWS CodePipeline, Docker, Kubernetes, Terraform (IaC), Maven, SonarQube, Git (GitHub, GitLab, Bitbucket).

Monitoring & Observability: Prometheus, Grafana, AWS CloudWatch, Log4j2, ELK Stack, Custom APM Dashboards, Alerting & Telemetry for Distributed Systems.

Security & Compliance: Role-Based Access Control (RBAC), OAuth 2.0, JWT, Encryption in Transit/At Rest, API Gateway Policies.

Development Practices: Agile/Scrum, CI/CD Pipelines, Test-Driven Development (TDD), Behavior-Driven Development (BDD), Cloud-Native Development, Microservices Lifecycle Management.

Tooling & Environments: IntelliJ IDEA, Visual Studio Code, Postman, Figma, JIRA, Confluence, Asana, Trello, Smartsheet.

Professional Attributes: Cross-Functional Collaboration, Technical Mentorship, Problem Solving, Documentation Excellence, Stakeholder Communication, Time Management, Quality-First Delivery.

Professional Experience

American Express

Aug 2024 – Present

Full Stack Developer

Scottsdale, AZ

- Architected and deployed **cloud-native microservices** using Spring Boot, Spring Cloud, and AWS Lambda for transaction management, achieving **40% lower API latency** and seamless multi-tenant scaling.
- Re-engineered **monolithic payment modules into distributed event-driven services**, leveraging Kafka and Redis queues to improve reliability and parallel request handling under 10K+ concurrent user loads.
- Implemented **retrieval-augmented generation (RAG)** pipelines combining Pinecone vector stores with LLaMA-3 fine-tuning for contextual insights in customer query resolution.
- Engineered and deployed **.NET 8 Web APIs in C#** to handle customer onboarding and transaction validation, integrating them with Azure SQL and microservices written in Spring Boot to achieve sub-200ms response latency in production.
- Built and productionized an **AI-driven anomaly detection engine** using Azure Machine Learning and TensorFlow that analyzed 10K+ daily transactions; integrated predictions directly into the .NET API layer for real-time fraud flagging and compliance alerts.
- Developed **React + TypeScript** dashboards with SSR and component-level caching, reducing first-paint times by **35%** while ensuring full WCAG 2.1 accessibility compliance.
- Designed **GraphQL schemas** using Netflix DGS and integrated schema stitching for cross-service data aggregation, cutting frontend over-fetch and reducing response payloads by **28%**.

- Automated CI/CD pipelines through **GitHub Actions + Jenkins**, implementing staged rollouts and monitoring with **Prometheus and Grafana** for microservice and ML model performance tracking.
- Collaborated with AI and DevOps teams to productionize **MLOps workflows** in AWS SageMaker, managing model retraining, versioning, and metric tracking using MLflow.
- Mentored junior developers and led internal workshops on **LangChain, RAG architectures, and full-stack API design**, improving team productivity and code quality by **20%**.
- Integrated **OpenAI APIs** within the .NET backend to extract entities, sentiments, and document insights from unstructured financial data, enabling automated report generation and reducing manual data-processing effort by 35%.

GSK

Aug 2019 – Jun 2022

Software Engineer

Bangalore, India

- Engineered and enhanced scalable backend services using **Java, Spring Boot, and AWS** to support marketing and analytics workflows, optimizing API reliability and data throughput across enterprise systems.
- Collaborated with cross-functional teams to build and integrate **RESTful APIs** for identity, campaign, and analytics modules, ensuring seamless data synchronization and reducing inter-service latency by **22%**.
- Built and deployed **.NET APIs integrated with Microsoft Dynamics 365 Customer Service**, automating ticket classification and improving SLA compliance by 25%.
- Customized **UI Portals** to display analytics dashboards and integrated workflows, enhancing visibility and self-service access for internal teams.
- Implemented unit and integration testing frameworks with **JUnit and Mockito**, increasing test coverage and reducing post-deployment bugs by **30%**.
- Optimized **Java microservices performance** through asynchronous event handling and load-balancing strategies, enhancing average response times by **15%** for data-intensive APIs.
- Architected and maintained **SQL and NoSQL databases** (PostgreSQL, MongoDB) for high-volume event and configuration data, improving query efficiency and supporting large-scale analytics pipelines.
- Containerized and deployed services using **Docker and Kubernetes**, enabling scalable multi-tenant microservice orchestration across production and staging environments.
- Developed internal web dashboards with **React.js and TypeScript** to monitor API performance, error logs, and campaign metrics, improving operational visibility for engineering and business teams.
- Implemented **WCAG 2.1 accessibility standards** and responsive design for healthcare analytics dashboards, focusing on inclusive and data-driven solutions that enhance both patient engagement and learning experiences for internal teams.
- Created internal automation scripts and monitoring tools for recurring maintenance tasks, boosting overall developer productivity by **20%**.

Education

Clemson University 2022 - 2024

CGPA 3.86/4

Master's of Science in Computer Science

Projects

DataOps Pipeline for Clinical Analytics | *Python, SQL, Airflow, AWS S3, Snowflake, dbt, Tableau*

- * Designed secure ETL workflows to ingest, cleanse, and transform multi-source clinical datasets into AWS S3 and Snowflake, ensuring accuracy and HIPAA compliance.
- * Implemented automated data validation using dbt tests and metadata monitoring, improving data reliability and reducing analytics delays by 45%.
- * Delivered self-service Tableau dashboards for cross-functional teams, accelerating decision-making and reducing manual reporting workload.

Stock Price Forecasting Platform | *LSTM, ARIMA, XGBoost, Azure ML, Airflow, Snowflake*

- * Developed an LSTM-based time-series forecasting model achieving 80–85% accuracy across benchmark equities, identifying short-term trend volatility.
- * Also Automated feature engineering, retraining, and model registry using Azure ML + Airflow pipelines, enhancing model reproducibility.
- * Benchmarked ARIMA and XGBoost baselines, improving forecasting accuracy by 10% through optimized hyperparameters and seasonal decomposition techniques.

Certification

AWS Certified Solutions Architect - Associate

Microsoft Azure Cloud - AZ900